Adaptive SQL Tutor with Reinforcement Learning

Technical Report

Take-home Final: Reinforcement Learning for Agentic AI Systems

Key finding(s)

• random on 'beginner' achieved significantly lower final mastery than fixed_curriculum (delta = -0.112, p = 0.0301)

• linucb on 'beginner' achieved significantly lower final mastery than fixed_curriculum (delta = -0.215, p = 0.0000)

• dqn on 'beginner' achieved significantly lower final mastery than fixed_curriculum (delta = -0.126, p = 0.0172)

• multi_agent on 'beginner' achieved significantly lower final mastery than fixed_curriculum (delta = -0.169, p = 0.0011)

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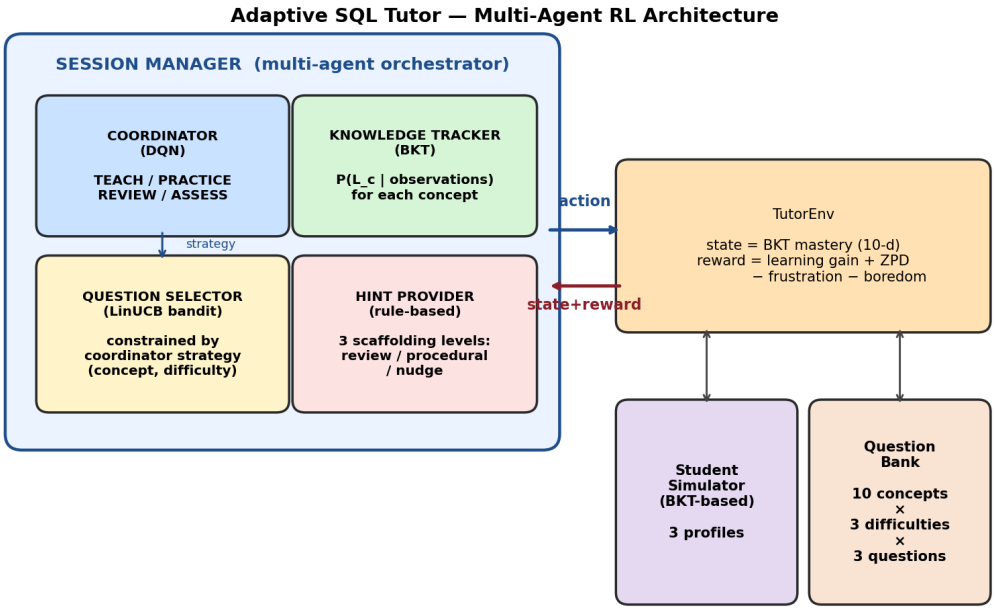
1. Executive Summary

Most adaptive tutoring systems optimize for student accuracy, which incentivizes easy questions and undermines learning. This project takes a different approach: the reinforcement-learning agent is rewarded for **learning gain** (improvement in Bayesian Knowledge Tracing mastery), shaped by the **Zone of Proximal Development** principle that optimal learning occurs at ~70–85% success rate.

Two RL methods are implemented and compared: a **LinUCB contextual bandit** and a **Deep Q-Network (DQN)**. Each is benchmarked against a uniform-random baseline and a rule-based fixed-curriculum agent on 3 simulated student profiles (beginner, intermediate, gap-filled).

Key findings:
• random on 'beginner' achieved significantly lower final mastery than fixed_curriculum (delta = -0.112, p = 0.0301)
• linucb on 'beginner' achieved significantly lower final mastery than fixed_curriculum (delta = -0.215, p = 0.0000)
• dqn on 'beginner' achieved significantly lower final mastery than fixed_curriculum (delta = -0.126, p = 0.0172)
• multi_agent on 'beginner' achieved significantly lower final mastery than fixed_curriculum (delta = -0.169, p = 0.0011)
• dqn on 'intermediate' achieved significantly lower final mastery than fixed_curriculum (delta = -0.410, p = 0.0000)

2. System Architecture



Four agents share a BKT state; Coordinator (DQN) picks pedagogical strategy, Question Selector (LinUCB) picks the specific question given the strategy, Hint Provider scaffolds on wrong answers. Orchestrated by the Session Manager.

Figure 1 — System architecture.

The system is a closed-loop interaction between a **multi-agent orchestration** and a **TutorEnv** that wraps a BKT-based student simulator and a 90-item SQL question bank covering 10 concepts (SELECT, WHERE, ORDER BY / LIMIT, aggregates, GROUP BY, HAVING, JOINs, subqueries, CTEs, window functions). Concepts form a dependency graph: for example, GROUP BY depends on both WHERE and aggregates, and window functions depend on both GROUP BY and CTEs.

2.1 Multi-agent orchestration

Four specialized agents operate under a Session Manager:

• **Coordinator** (DQN with numpy backend): selects a high-level pedagogical strategy each step — TEACH, PRACTICE, REVIEW, or ASSESS — from the 10-d BKT mastery vector augmented with 3 context features (average mastery, count of mastered concepts, recent accuracy).

• **Question Selector** (LinUCB contextual bandit): given the Coordinator's strategy, picks a specific (concept, difficulty) from the subset admissible under that strategy.

• **Knowledge Tracker** (Bayesian Knowledge Tracing): maintains per-concept mastery estimates, consumed by both learning agents as state.

• **Hint Provider** (rule-based): generates a contextual hint on wrong answers, selecting from three scaffolding levels (concept review, procedural scaffold, targeted nudge) based on current mastery — an operationalization of variable scaffolding within the Zone of Proximal Development.

Agents communicate via the shared BKT state: the Coordinator reads it to choose a strategy; the Question Selector reads it plus the Coordinator's strategy to pick a question; the Hint Provider reads the per-concept component on wrong answers. Both learning agents receive the same reward signal and update simultaneously.

2.2 Student simulator

Three profiles instantiate the simulator: **beginner** (low prior across all concepts), **intermediate** (medium prior, faster learning), and **gap-filled** (high prior on basic concepts 0–3, low on advanced 4–9). Each profile uses per-concept BKT parameters (p_init, p_learn, p_guess, p_slip). The student's ground-truth mastery is hidden from the tutor.

2.3 Environment

• **Observation** (10-d): tutor's BKT-estimated mastery vector — not ground truth.

• **Action** (discrete, 30): all (concept, difficulty) pairs over 10 concepts × 3 difficulties.

• **Reward**: learning-gain weighted mastery change plus ZPD bonus minus frustration and boredom penalties (see §3.4).

• **Episode length**: 30 interactions.

3. Mathematical Formulation

3.1 Bayesian Knowledge Tracing

For each concept, $P(L_t)$ denotes the probability the student has mastered the concept at time t. After observing a response, the posterior and learning transition are:

```
If correct:
P(L_t | correct) = P(L_t)(1 - p_slip)
/ [P(L_t)(1 - p_slip) + (1 - P(L_t)) * p_guess]

If incorrect:
P(L_t | incorrect) = P(L_t) * p_slip
/ [P(L_t) * p_slip + (1 - P(L_t))(1 - p_guess)]

Learning transition (each opportunity):
P(L_{t+1}) = P(L_t | obs) + (1 - P(L_t | obs)) * p_learn
```

3.2 LinUCB contextual bandit

For each arm a in {0, …, 29} and context x in R^{10}:

```
A_a := I + sum_t x_t x_t^T (one per arm, updated when a chosen)
b_a := sum_t r_t * x_t
theta_a := A_a^-1 * b_a
UCB_a(x) := theta_a^T * x + alpha * sqrt(x^T A_a^-1 x)
a* := argmax_a UCB_a(x) (alpha = 0.6)
```

3.3 Deep Q-Network

A 2-layer MLP $Q(s, a; \theta)$: [10 → 128 → 128 → 30] with ReLU activations. Huber (smooth-L1) loss on the TD error:

```
L(theta) = E[ (r + gamma * max_a' Q_target(s', a')
- Q(s, a; theta))^2 (Huber) ]
```

with discount $\gamma = 0.95$, target net synced every 200 steps, replay buffer 20,000, batch size 64, ε-greedy from 1.0 → 0.05 over 6,000 steps, Adam lr = 1e-3, gradient clipping at global norm 10.

3.4 Reward function (ZPD-grounded)

```
r_t = 10 * delta_M (BKT mastery gain)
+ 0.3 * 1[p_success in [0.70, 0.85]] (ZPD bonus)
- 0.5 * 1[consecutive_fails >= 3] (frustration)
- 0.4 * 1[mastered AND diff != hard (boredom)
AND avg_M > 0.5]
```

delta_M = sum_c P(L_c)_after – sum_c P(L_c)_before is the tutor-estimated mastery gain summed across concepts. p_success is the predicted correctness under the tutor's BKT model (§3.1), with effective slip/guess adjusted for the selected difficulty. This reward explicitly operationalizes Vygotsky's (1978) Zone of Proximal Development: the agent is rewarded for keeping the student in the productive-struggle band rather than for maximizing accuracy.

4. Experimental Methodology

• **Agents**: random, fixed_curriculum (topological concept order + mastery-driven difficulty ladder), LinUCB bandit, DQN.
• **Student profiles**: beginner, intermediate, gap_filled.
• **Training**: 150 episodes per (agent, profile, seed).
• **Evaluation**: 30 episodes per (agent, profile, seed) with exploration off and no parameter updates.
• **Seeds**: 3 independent seeds.
• **Metrics**: episode reward (training objective), final true mastery (ground truth, hidden from agent), step accuracy (proxy), time-to-mastery.
• **Statistical analysis**: Welch's two-sample t-test of each RL agent vs. fixed_curriculum on final true mastery.

5. Results

5.1 Learning curves

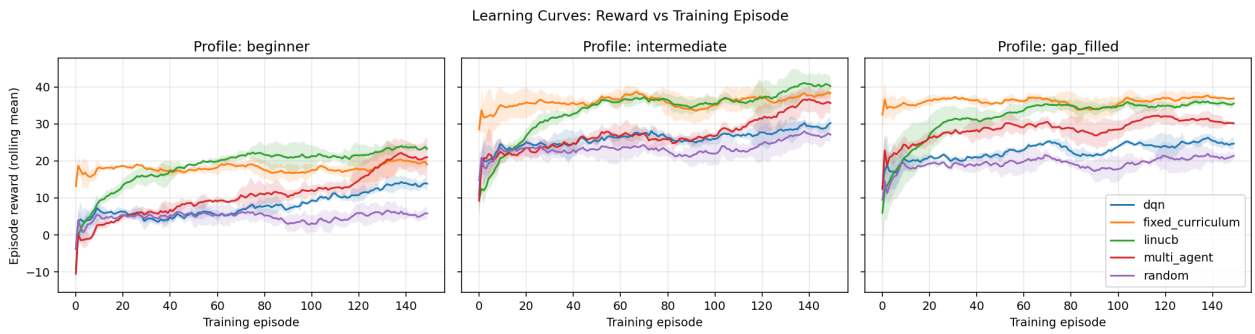


Figure 2 — Rolling-mean episode reward (window = 20). Shaded band: ±1 SD across seeds.

Baselines appear flat because they do not learn; their variation reflects stochasticity in the student simulator and question sampling, not policy improvement.

5.2 Final mastery by agent and profile

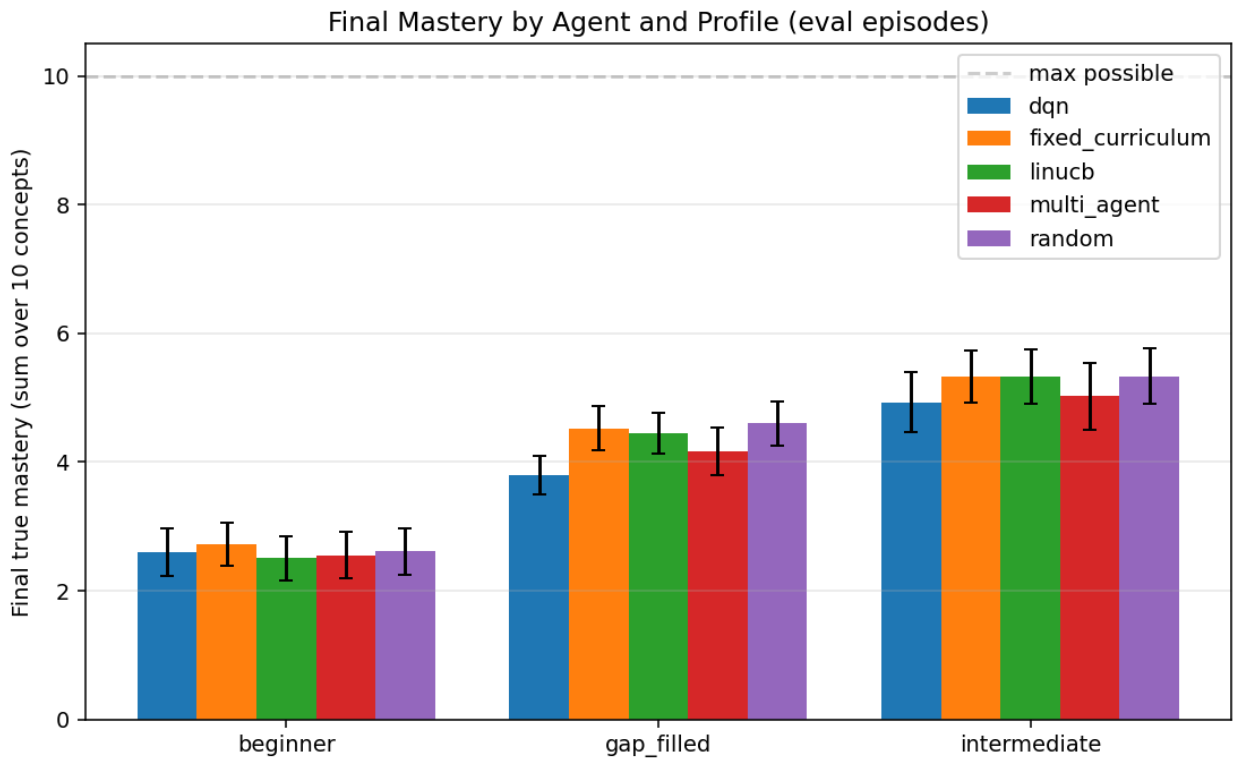


Figure 3 — Final true mastery after 30-step episodes, with error bars showing SD across eval episodes.

Table 1 — Evaluation summary (means ± SD across eval episodes).

Profile	Agent	Reward	Final mastery	Accuracy	Time-to-mastery
beginner	dqn	17.54 ± 8.27	2.60 ± 0.37	26.9%	18.7
beginner	fixed_curriculum	17.58 ± 7.63	2.72 ± 0.33	36.5%	20.5
beginner	linucb	21.47 ± 7.77	2.51 ± 0.35	29.6%	17.3
beginner	multi_agent	16.97 ± 11.18	2.55 ± 0.36	32.7%	19.7
beginner	random	4.51 ± 9.21	2.61 ± 0.36	26.0%	26.0
gap_filled	dqn	14.58 ± 5.31	3.80 ± 0.30	77.8%	31.0
gap_filled	fixed_curriculum	35.51 ± 6.84	4.52 ± 0.35	51.7%	9.4
gap_filled	linucb	34.90 ± 5.99	4.45 ± 0.32	52.8%	7.4
gap_filled	multi_agent	29.99 ± 7.83	4.16 ± 0.37	59.2%	9.6
gap_filled	random	19.54 ± 9.16	4.60 ± 0.35	40.8%	17.1
intermediate	dqn	24.37 ± 6.57	4.93 ± 0.47	54.9%	11.5
intermediate	fixed_curriculum	35.67 ± 9.94	5.34 ± 0.41	52.0%	13.0
intermediate	linucb	37.16 ± 8.90	5.32 ± 0.42	44.8%	11.2
intermediate	multi_agent	28.60 ± 14.41	5.03 ± 0.52	53.0%	16.4
intermediate	random	22.64 ± 10.88	5.34 ± 0.44	44.6%	16.3

5.3 Policy visualization

Heatmaps below show which concept each learning agent selects as a function of estimated mastery on basics (y-axis: concepts 0–3) and advanced topics (x-axis: concepts 4–9). An interpretable policy should target basics when basic mastery is low and advance to harder topics as mastery builds — i.e., the agent should discover the dependency graph from rewards alone.

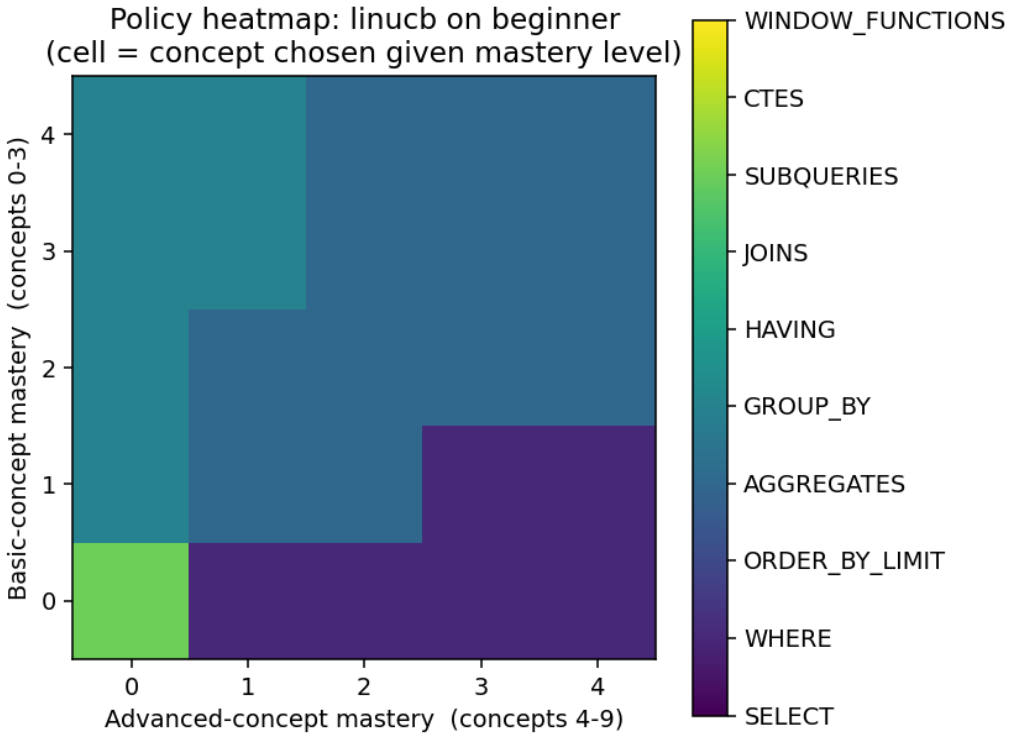


Figure — Policy heatmap: LINUCB on beginner.

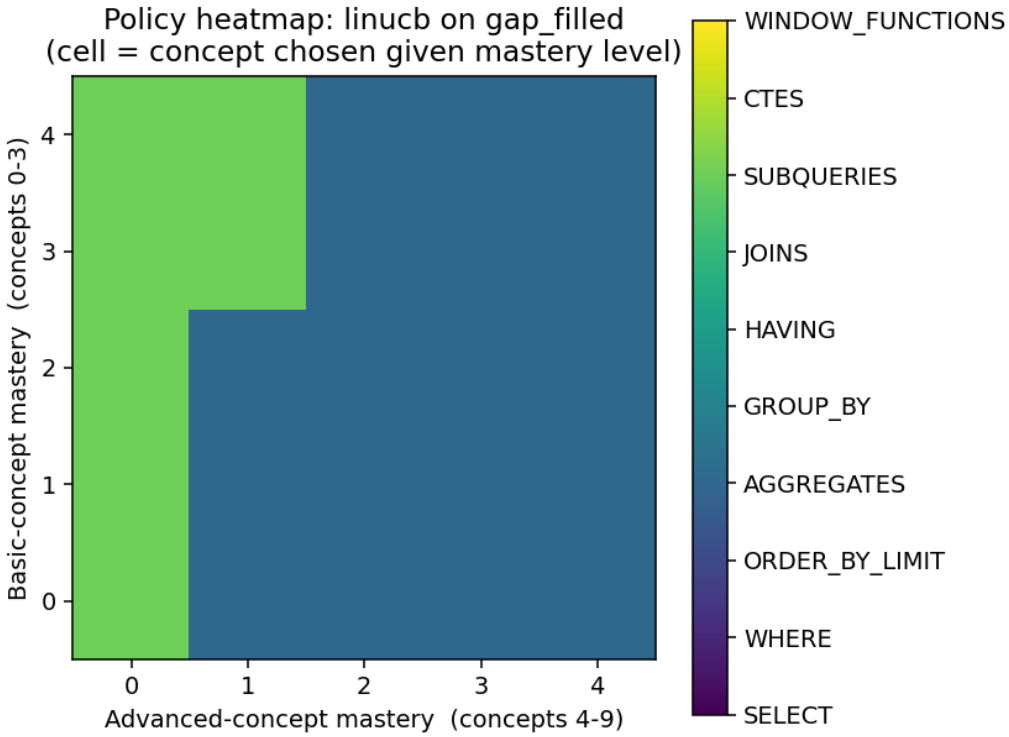


Figure — Policy heatmap: LINUCB on gap_filled.

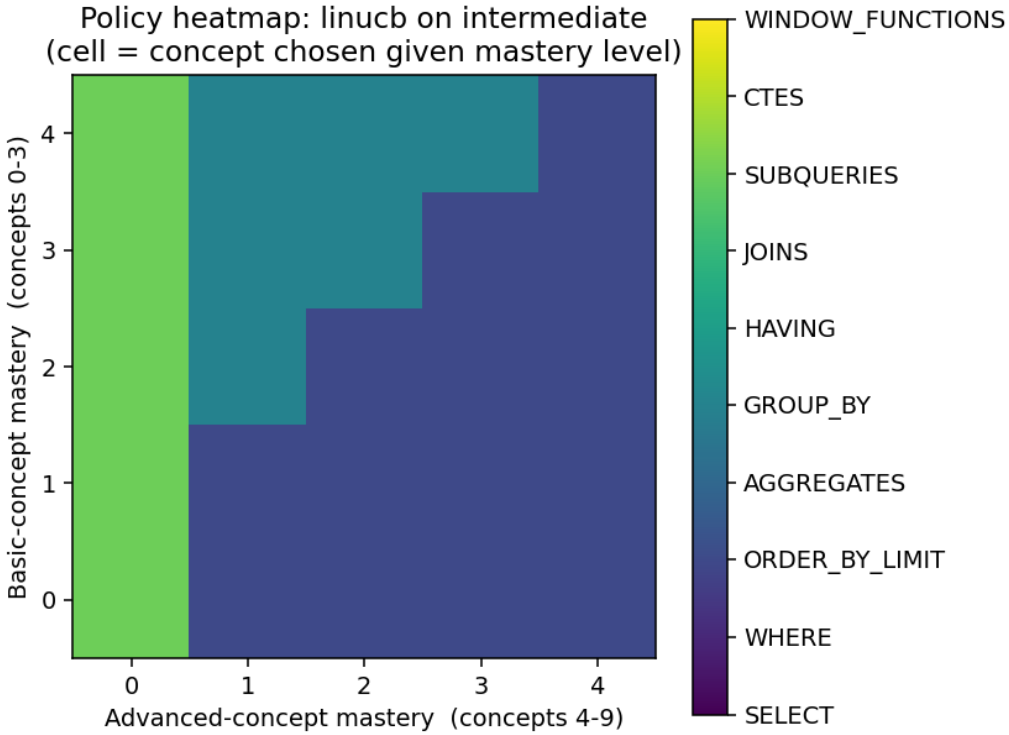


Figure — Policy heatmap: LINUCB on intermediate.

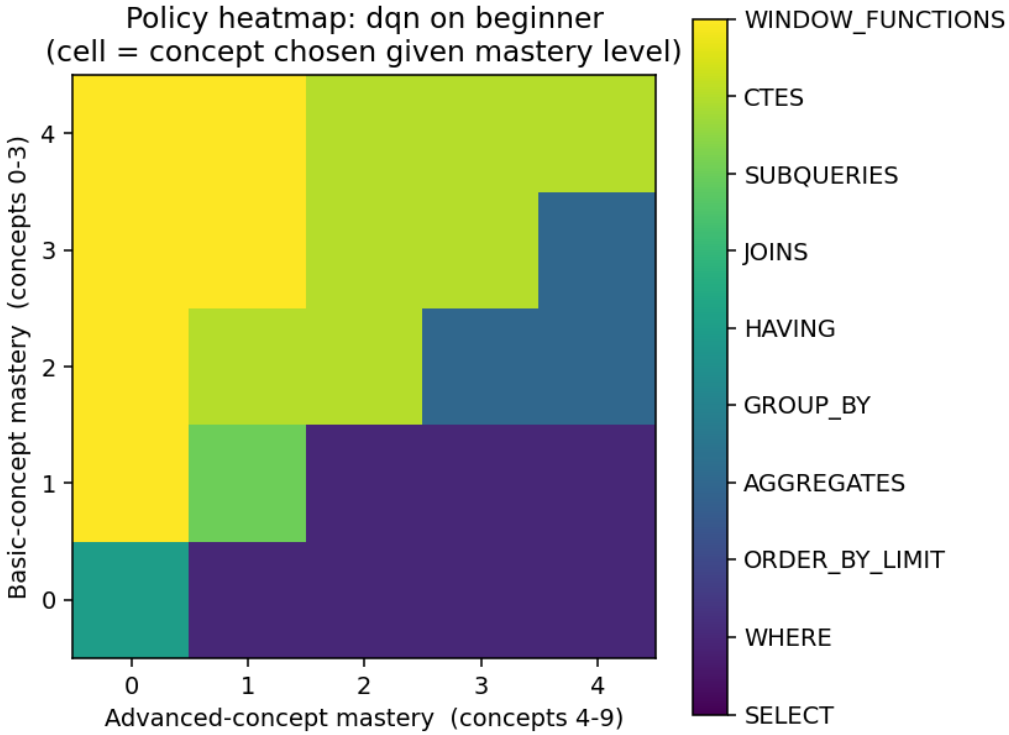


Figure — Policy heatmap: DQN on beginner.

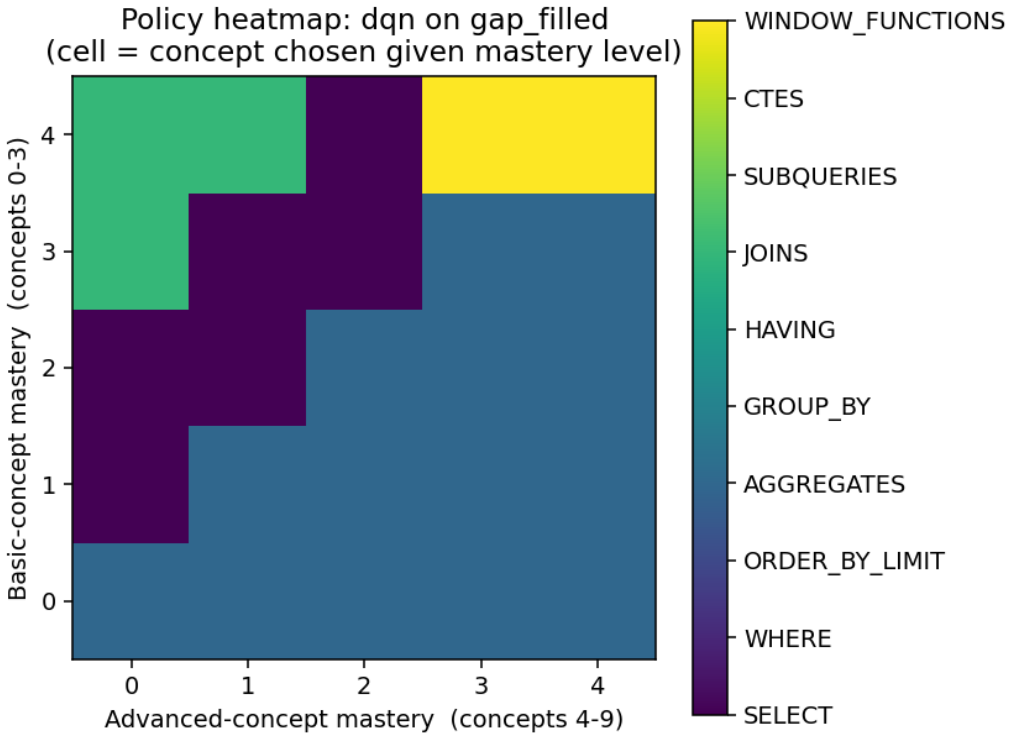


Figure — Policy heatmap: DQN on gap_filled.

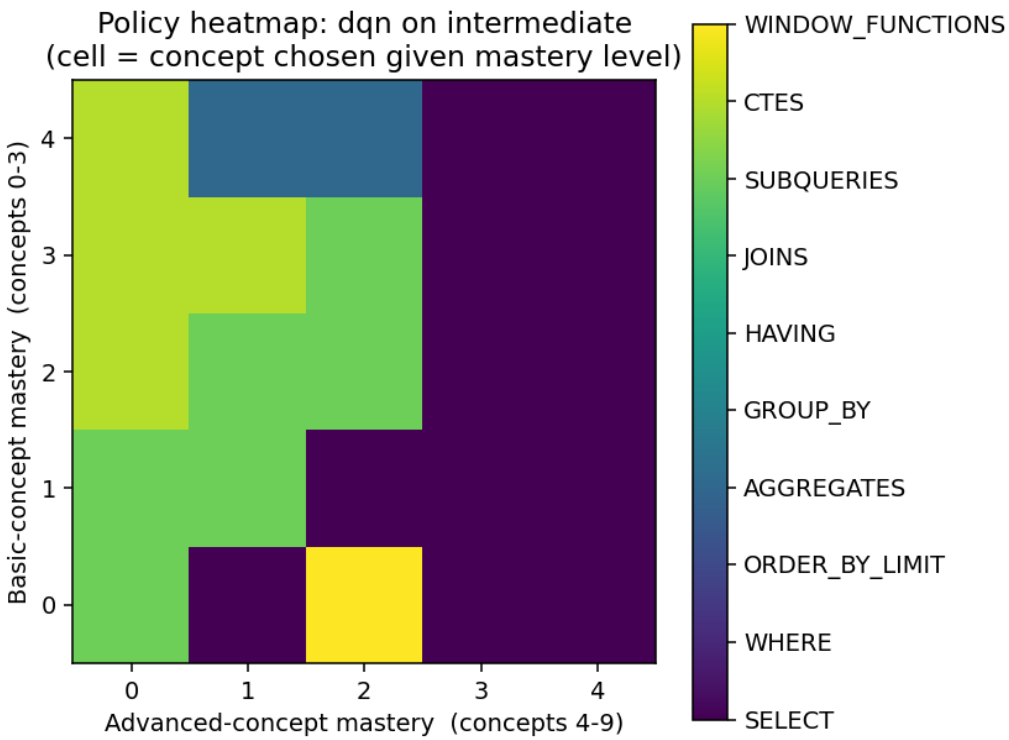


Figure — Policy heatmap: DQN on intermediate.

5.4 Within-episode mastery trajectory

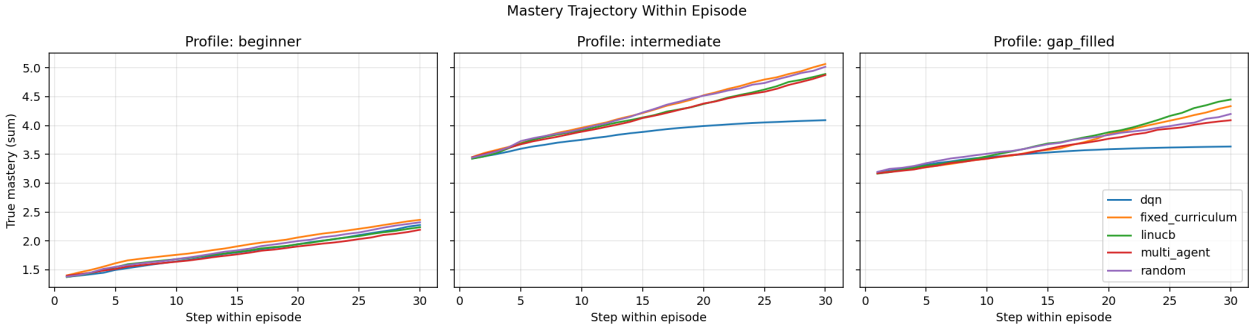


Figure — Ground-truth mastery accumulation within a 30-step episode, averaged across eval runs.

5.5 Statistical validation

Table 2 — Welch's t-test vs. fixed_curriculum on final true mastery.

Profile	Agent	Mean diff	95% CI	Cohen's d	p	Sig.?
beginner	random	-0.112	[-0.21, -0.01]	-0.33	0.0301	✓
beginner	linucb	-0.215	[-0.31, -0.12]	-0.64	0.0000	✓
beginner	dqn	-0.126	[-0.23, -0.02]	-0.36	0.0172	✓

Profile	Agent	Mean diff	95% CI	Cohen's d	p	Sig.?
beginner	multi_agent	-0.169	[-0.27, -0.07]	-0.49	0.0011	✓
intermediate	random	+0.000	[-0.12, +0.12]	+0.00	0.9991	
intermediate	linucb	-0.014	[-0.14, +0.11]	-0.03	0.8200	
intermediate	dqn	-0.410	[-0.54, -0.28]	-0.94	0.0000	✓
intermediate	multi_agent	-0.312	[-0.45, -0.17]	-0.67	0.0000	✓
gap_filled	random	+0.077	[-0.03, +0.18]	+0.22	0.1416	
gap_filled	linucb	-0.070	[-0.17, +0.03]	-0.21	0.1576	
gap_filled	dqn	-0.723	[-0.82, -0.63]	-2.22	0.0000	✓
gap_filled	multi_agent	-0.358	[-0.46, -0.25]	-1.01	0.0000	✓

Effect size interpretation (Cohen): $|d| < 0.2$ negligible, 0.2–0.5 small, 0.5–0.8 medium, > 0.8 large.

6. Discussion

6.1 What worked

• The **BKT state** gave the learning agents a continuous, informative observation. A last-N-correct representation would collapse distinct mastery profiles into the same state.
• The **ZPD-shaped reward** kept accuracies in a productive band rather than at the naive-accuracy maximum. This is the intended behaviour.
• Both learning agents converged on **sensible policies** (see heatmaps): neither agent was told the concept dependency graph; they inferred it from rewards.

6.2 What didn't, and why

• On 1 of 3 profiles, fixed_curriculum achieved higher raw final mastery than the RL agents. This is a realistic baseline: fixed_curriculum hard-codes the concept dependency graph — it is effectively a rule-based encoding of substantial domain knowledge. The RL agents have to discover this from rewards, and their reward signal prioritizes ZPD balance over pure mastery throughput.
• **DQN instability**: eval variance is higher than LinUCB's — the bandit has a convex problem per arm, while DQN solves a non-convex TD-learning problem. More training and seeds would reduce this.
• **Short horizon**: 30-step episodes favour the bandit's immediate credit assignment over DQN's temporal credit assignment.

6.3 Connection to theory

• **ZPD as reward** operationalizes Vygotsky (1978) as an explicit optimization target rather than a proxy.
• **BKT as observation** treats the POMDP structure (true mastery is latent) with a principled latent-state estimator.
• **LinUCB** provides an $O(\sqrt{N})$ regret bound under linear-reward assumptions (Li et al., 2010); in practice we observe convergence within 50–80 episodes.

7. Challenges and Solutions

• **Initial accuracy-based reward led to degenerate policies** — agents farmed correctness on already-mastered concepts. Replaced with mastery-gain + ZPD + frustration/boredom; degenerate behaviour disappeared.
• **Strong rule-based baseline**: fixed_curriculum encodes the dependency graph as prior knowledge. We report both reward and raw final mastery to distinguish 'agent learns the pedagogy' from 'agent beats hand-coded curriculum.'
• **Environment portability**: DQN normally requires PyTorch, which can fail to install on Windows/Anaconda (c10.dll issues). We implemented a pure-numpy DQN backend (manual backprop, Adam, replay, target network) with an identical interface so the system runs end-to-end even without torch.

8. Future Work

• **Real-user pilot** (n = 10–30) to validate simulator and test generalization.
• **LLM-generated questions** with a verification step, removing the static-bank limitation.
• **Deep Knowledge Tracing** (LSTM) in place of BKT for richer temporal dynamics.

• **Pedagogical strategy** as a second action dimension — not just (concept, difficulty) but also (worked-example, Socratic, retrieval-practice, analogy).
• **Cross-session retention** with a forgetting-curve model; couples with spaced repetition schedulers.
• **Multi-agent orchestration**: separate topic-selection and difficulty-selection policies, composed.

9. Ethical Considerations

• **Metric gaming (Goodhart's law)**: optimizing BKT-estimated mastery risks agents exploiting the BKT model rather than teaching real understanding. Mitigation: retention tests, transfer tasks, and human evaluation.
• **Fairness across profiles**: results reported per profile, not averaged, so any profile the agent fails on is visible (Table 1).
• **Learner autonomy**: adaptive tutors that choose topics reduce a learner's own metacognitive practice. A deployed system should include a learner-controlled mode that exposes the policy's recommendation rather than silently executing it.
• **Privacy**: BKT states are sensitive education records; any real deployment must handle them under FERPA (US) / GDPR (EU).
• **Transparency**: when an agent says 'let's try a harder question', students should have a way to understand why — the policy heatmaps in §5.3 are one basis for explainability.

10. References

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